Vikash Vikash

Department of Chemistry and Chemical Engineering
Division of Chemical Engineering
Chalmers University of Technology
412 96 Gothenburg, Sweden
vikasche.engg@gmail.com; vikvik@chalmers.se
ORCiD: 0000-0003-4644-2624

Academic Appointments

12/2025 – present
Researcher, Department of Chemistry and Chemical Engineering
Division of Chemical Engineering
Chalmers University of Technology, Gothenburg, Sweden.
- Research focuses on multiphase flow, droplet- and bubble-laden turbulent flows and interfacial dynamics and multiphysics CFD
- Collaborations: NitroCapt AB (Sweden); Maastricht University (Netherlands).

12/2022 – 12/2025
Postdoctoral Fellow, Department of Chemistry and Chemical Engineering
Division of Chemical Engineering
Chalmers University of Technology, Gothenburg, Sweden.
- Developed high-fidelity modelling workflows for multiphase systems of droplets and bubbles in turbulent flow conditions and 3D CFD models development for microwave plasma reactor
- Collaborations: NitroCapt AB (Sweden); Maastricht University (Netherlands).

01/2022 – 04/2022
Research Associate, Department of Chemical Engineering
Indian Institute of Technology Guwahati, India.
- Research focused on multiphase modeling of shear-thinning rivulet flow on inclined surfaces
- Collaborations: Reckitt Benckiser India.

Education

2021 –
PhD in Chemical Engineering
Indian Institute of Technology Roorkee, India.

2013 –
MTech in Chemical Engineering
Indian Institute of Technology Roorkee, India.

2010 –
Bachelor of Engineering [BE] in Chemical Engineering
Maharishi Dayanand University, Rohtak, India.

Publications [selected]

2026 –
Vashisth, V., Iqbal M., Butterworth T., van Rooij G., Andersson R., Analysis of forced and intrinsic quenching in a plasma reactor for nitrogen fixation by use of three-dimensional simulations and experiments, *Chemical Engineering Journal*, Vol. 543, 178575.

Description: Lead author. Developed validated 3D high-resolution CFD framework to quantify forced jet-quenching effects on thermo-fluid-chemical dynamics of NO_x production in electrified microwave-plasma for nitrogen fixation.

2025 – **Vashisth V.**, Andersson R., Dynamics of bubble breakup under turbulent flow conditions, *Chemical Engineering Journal*, Vol. 511, 162019.

Description: Lead author. Developed and validated modelling framework capturing stochastic bubble breakup, resolving internal flow mechanisms and providing benchmark datasets to improve breakup models in turbulent multiphase operations.

2024 – Tatar M., **Vashisth V.**, Iqbal M., Butterworth T., van Rooij G., Andersson R., Analysis of a plasma reactor performance for direct nitrogen fixation by use of three-dimensional simulations and experiments, *Chemical Engineering Journal*, Vol. 497, 154756.

Description: Co-author. Developed a validated 3D CFD-chemistry coupled modelling framework for microwave-plasma nitrogen-fixation and sustainable production.

2022 – **Vashisth V.**, Kumar V., Effect of stator geometries on flow fields and mixing performance for viscous fluids, *Chemical Engineering and Processing: Process Intensification*, Vol. 180, 108595.

Description: Lead author. Developed a methodology for rotor–stator mixer to quantify the influence of stator design and flow parameters on hydrodynamics and mixing metrics for better emulsification and de-agglomeration.

2022 – **Vashisth V.**, Nigam K.D.P., Kumar V., Nanoparticle de-agglomeration in viscous fluids using different high shear mixer geometries. *Chemical Engineering Science*, Vol. 248 Part A, 117132.

Description: Lead and corresponding author. Quantified how rotor–stator geometry and viscosity govern nanoparticle de-agglomeration identifying erosion-dominated breakup and providing design-relevant guidance for scalable dispersion processes.

2021 – **Vashisth V.**, Nigam K.D.P., Kumar V., Design and development of high shear mixers: fundamentals, applications and recent progress, *Chemical Engineering Science*, Vol. 232, 116296.

Description: Lead author. Connecting design to hydrodynamics and drop/particle size control, and positioning CFD as a digitalised route toward predictive design and scale-up in advanced manufacturing.

2020 – **Vikash**, Kumar V., Ultrasonic-assisted de-agglomeration and power draw characterization of silica nanoparticles, *Ultrasonics Sonochemistry*, Vol. 65, 105061.

Description: Lead author. Utilised energy-intensive dispersion process for nanoparticle clusters and established breakup mechanisms. The resulting process supports a central step to sustainable manufacturing.

2019 – **Vikash**, Kumar V., Turbulent statistics of flow fields using large eddy simulations in batch high shear mixers, *Chemical Engineering Research and Design*, Vol. 147, 561–569.

Description: Lead author. Performed the first high-resolution simulation study of batch rotor–stator mixers, validating power consumption and quantifying hydrodynamics and turbulence statistics.

2019 – Kushwaha, N., **Vikash**, Kumar, V., Impact of mixed convective and radiative heat transfer in spiral coiled-tubes, *Journal of Heat Transfer ASME*, Vol. 141, 081001–10.

Description: Co-author. A CFD study coupling turbulent convection with thermal radiation modelling in spiral-coiled tubes to enhance heat transfer (e.g., exhaust heat recovery, solar thermal systems).

2017 – **Vikash**, Deshawar D., Kumar V., Hydrodynamics and mixing characterization in a novel high shear mixer, *Chemical Engineering and Processing: Process Intensification*, Vol. 120, 57–67.

Description: Lead author. Developed a novel batch rotor–stator mixer configuration supporting digitally enabled process intensification and scale-up relevant to sustainable manufacturing.

Total publications – 15+ [Journals and Conferences] of which 2 are in preparation. Citations: 379; h-index: 10 (as of 24 August 2026) [Google Scholar].

Industry and Academic Collaborations

NitroCapt AB (Sweden) – Industrial collaboration during Postdoc/Researcher period at Chalmers University of Technology, Sweden.

Maastricht University (Netherlands) – Academic collaboration during Postdoc/Researcher period at Chalmers University of Technology, Sweden.

Loughborough University (UK) – Academic collaboration with School of Aeronautical, Automotive, Chemical and Materials Engineering during Institute of Advanced Studies Residential Fellowship.

Reckitt Benckiser (India) – Industrial collaboration during Research Associate period at Indian Institute of Technology Guwahati, India.

Indian Institute of Technology Delhi – Academic collaboration during PhD at Indian Institute of Technology Roorkee, India.

Grants, Awards and Fellowships

2022 – Institute of Advanced Studies Residential Fellowship
Fellowship open to outstanding international researchers from across all disciplines and career stages by the Loughborough University, UK. Researched at School of Aeronautical, Automotive, Chemical and Materials Engineering, Department of Chemical Engineering.

2020 – International Travel Grant
Indian Institute of Technology Roorkee, India.
A travel grant offered to outstanding students for scientific contribution and presenting research at international conferences or symposiums.

2015 – 2020 National Fellowship for PhD
Ministry of Education, India.
Fellowship for PhD studies offered by the Ministry of Education India at prestigious institutions across India.

2018 – International Travel Grant
Science and Engineering Research Board [SERB- DST], India.
Highly competitive travel grant offered to outstanding researchers/students for scientific contribution and presenting research at international conferences or symposiums.

2018 – International Travel Grant
Indian Institute of Technology Roorkee, India.
A travel grant offered to outstanding students for scientific contribution and presenting research at international conferences or symposiums.

2011 – 2013 National Fellowship for MTech.
Ministry of Education, India.
Fellowship for master's studies offered by the Ministry of Education India at prestigious institutions across India.

2012 – All India Rank – 494
Chemical Engineering
Graduate Aptitude Test in Engineering [GATE].

2010 – All India Rank – 55
Chemical Engineering
Graduate Aptitude Test in Engineering [GATE].

Invited Talks and Selected Presentations

06/2024 – Mechanistic study on bubbles–droplets–vortices interaction and breakup in turbulent flows, *International Symposium on Chemical Reaction Engineering (ISCRE 28)*, Turku, Finland.

03/2023 – Process intensification through high shear mixing,
Institute of Advanced Studies, Loughborough University, UK.

11/2021 – Effect of tracer and fluid behavior on mixing performance of high shear mixers, *North American Mixing Forum*, AIChE Annual Meeting, Virtual.

11/2020 – Nanoparticle suspension in the viscous fluid using high shear mixing, *North American Mixing Forum*, AIChE Annual Meeting, Virtual.

12/2020 – High shear mixing for viscous fluids, *International Conference on Complex Fluids [COMPFLU-2020]*, IIT Bombay, India.

09/2018 – Mixing characterization in batch rotor–stator mixer, *16th European Conference on Mixing*, Toulouse, France.

11/2013 – Hydrodynamics and mixing characterization in a high shear mixer, *AIChE Annual Meeting*, San Francisco, USA.

Teaching

2019 – 2020
Tutorial instructor of Bachelor student's theory courses
Indian Institute of Technology Roorkee, India.
[~ 60 students]
[8 hours per week]
Responsibility: Tutorials; grading; supervision.

2016 – 2018
Experiment instructor of Bachelor student's laboratory courses
Indian Institute of Technology Roorkee, India.
[~ 40 students]
[8 hours per week]
Responsibility: Designed lab modules; grading; introduced laboratory equipment.

Pedagogical Training and Professional Development

2025 – 2026
Pedagogical Swedish language course: A2 Level completed, and planned with B1 Level in June 2026
Language Course offered by Upgrades
Chalmers University of Technology, Sweden.

2024 –
Short term course on Chemical Process Engineering
Department of Chemistry and Chemical Engineering
Chalmers University of Technology, Sweden.

2018 –
Characterization of Nanoparticles using Zetasizer: Hands on Training,
Centre of Nanotechnology
Indian Institute of Technology Roorkee, India.

2016 –
Short Term Course on Chemical Micro Process Engineering: Flow Modelling and Applications
Department of Chemical Engineering
Indian Institute of Technology Roorkee, India.

2016 – 2020
Pedagogical training through Active Learning and Lab-Based Learning
Department of Chemical Engineering
Indian Institute of Technology Roorkee, India.

2015 – 2016
University pedagogy course (PhD) – 14 Credits
Department of Chemical Engineering
Indian Institute of Technology Roorkee, India.

Supervision and Mentoring

2016 – 2017
Co-supervised MTech thesis: Fluid flow and heat transfer characteristics in curved tube.
Department of Chemical Engineering
Indian Institute of Technology Roorkee, India.

Academic Services and Memberships

Peer reviewer – Chemical Engineering Journal; Chemical Engineering Science; Chemical Engineering and Processing – Process Intensification; Industrial & Engineering Chemistry Research (ACS Journal); Results in Engineering.

Committee member – Department Research Committee, Department of Chemical Engineering, Indian Institute of Technology Roorkee, India [2016].

Conferences/Short courses – 12th International Conference on Complex Fluids and Soft Matter, [2018]; Short course on Chemical Micro Process Engineering: Flow Modelling and Applications, [2016]; International Conference on Advances in Chemical Engineering, Indian Institute of Technology Roorkee, India [2013].

Life associate member – Indian Institute of Chemical Engineers (IIChE)
Membership No.: LAM–70012.

References

Prof. Ronnie Andersson – Professor of Chemical Engineering and Director Centre for Chemical Process Engineering
Department of Chemistry and Chemical Engineering
Chalmers University of Technology, Sweden.
Email: ronnie.andersson@chalmers.se | Phone: 031-772 29 41

Prof. Vimal Kumar – Professor of Chemical Engineering
Department of Chemical Engineering
Indian Institute of Technology Roorkee, India.
Email: vimal.kumar@ch.iitr.ac.in | Phone: +91-1332 285694